Arunabha Majumder

+45 71 62 24 00 | Aalborg, Denmark
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder

November 4, 2026

Ottobock SE & Co. KGaA
Attn: Katharina Nolte

Open Application, Professional Talent Pool (Job ID 8362)

Dear Ms Nolte,

I would like to join Ottobock's talent pool. I am a mechanical and mechatronics engineer doing my PhD in Mechatronics and Robotics at Aalborg University, with my thesis expected to be finished in March. Most of my work has been on devices worn on the human body, so prosthetics, orthotics and exoskeletons at Ottobock are exactly where I want to build my career.

In my PhD I designed a hybrid shoulder exoskeleton where a spring carries most of the arm load and a small 6 Nm motor adds the rest. I took it from hand sketches to SolidWorks, optimised the design with a genetic algorithm, integrated the motor, encoder and force sensors, measured its stiffness on a physical test setup and tested it with 12 participants. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. In my Master's I designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive a pneumatic muscle. My first exoskeleton was a rehabilitation device for a paralysed arm in my Bachelor's.

Areas of interest: mechanical design and actuator development for exoskeletons, orthoses and prostheses, prototyping, and test and validation with users.

Preferred location: Duderstadt, as the centre of Ottobock's research and development. I am happy to relocate within Germany.

Notice period and availability: my thesis is expected to be finished in March, so I can start from spring 2027. I am available full-time.

I also hold a Senior Diploma in Fine Arts, so I often explain ideas with quick hand sketches, which helps when working with clinicians and users. Thank you for considering my application.

Kind regards,
Arunabha Majumder